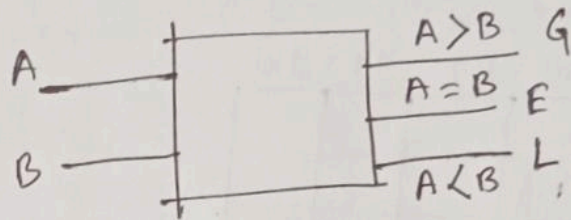
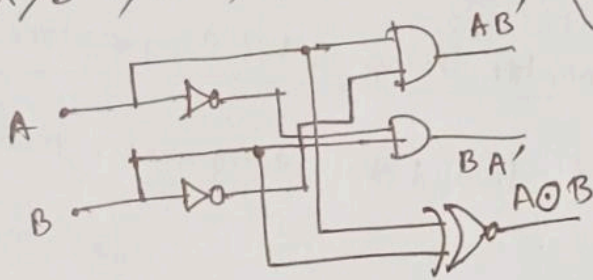


1 bit digital Magnitude Comparator

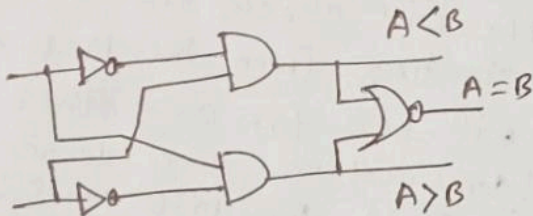
Inputs		$A > B$	$A = B$	$A < B$
A	B	G	E	L
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0



$A > B \Rightarrow AB'$, $A = B \Rightarrow AB + A'B'$, $A < B \Rightarrow A'B$
 APPLY NOR
 $(A'B + AB') \leftarrow \text{NOR}$



$= (A+B')(A'+B)$
 $= AA' + AB + A'B' + BB'$
 $= AB + A'B'$



2 bit Magnitude Comparator using TT & KMap

A_2	A_1	B_2	B_1	$A > B$	$A = B$	$A < B$
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	0	1	0
1	1	1	0	0	0	1
1	1	1	1	0	1	0

$A_2 A_1$		$B_2 B_1$	00	01	11	10
$B_2 B_1$	00			1	1	1
	01				1	1
	11					
	10					1
	00					

for $A > B = A_2 B_2' + A_1 A_2 B_1' + A_1 B_1' B_2'$

Kmap for $A = B$

$A_2 A_1$		$B_2 B_1$	00	01	11	10
$B_2 B_1$	00	1				
	01		1			
	11				1	
	10					1
	00					

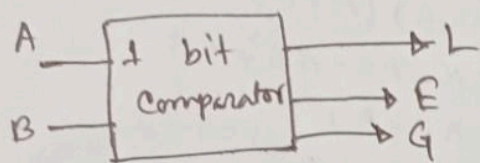
$(A_2 \odot B_2)(A_1 \odot B_1)$

K Map for $A < B$

$A_2 A_1$	00	01	11	10
$B_2 B_1$				
00				
01	1			
11	1	1		1
10	1	1		

$$= A_2' B_2 + A_1' A_2' B_1 + A_1' B_1 B_2$$

2 bit Comparator Using 1 bit Comparators



We need two 1 bit comparator module.

Let $A = A_1 A_0$, $B = B_1 B_0$
 First let's check for $A > B$.
 For $A > B$ We will check the condition starting from the MSB and will see for any other possible combination can come along with the MSB condition

$$A_1 \text{ should be greater than } B_1 \Rightarrow A_1 > B_1$$

or A_1 might be equal to B_1

$A_1 = B_1$ in that case, with $A_1 = B_1$, A_0 must be greater than B_0 otherwise our prime condition $A > B$ will not be valid.

So, the condition will be $A_1 > B_1$ or $A_1 = B_1$ and $A_0 > B_0$

So, for final G (greater) condition

$$G = G_1 + E_1 \cdot G_0$$

Now for $A = B$

For this Both the digit must be equal

$$A_1 = B_1 \text{ and } A_0 = B_0$$

$$E = E_1 \cdot E_0$$

Now for $A < B$

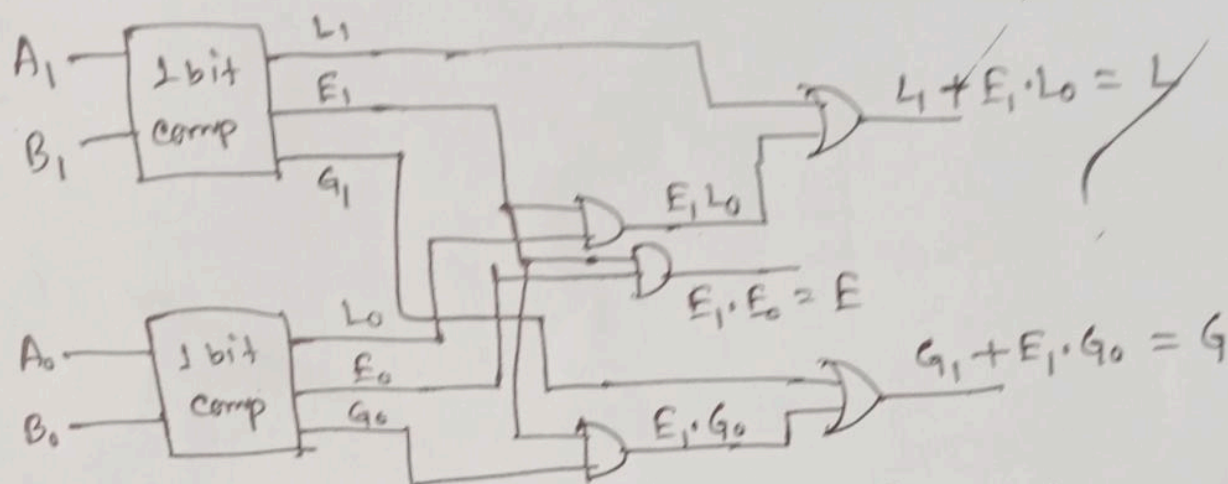
A_1 must be less than $B_1 \Rightarrow A_1 < B_1$

or $A_1 = B_1$ (A_1 might be equal to B_1) if $A_1 = B_1$, then

A_0 must be less than B_0

$$A < B \Rightarrow A_1 < B_1 \text{ or } A_1 = B_1 \text{ and } A_0 < B_0$$

$$L = L_1 + E_1 \cdot L_0$$



Similarly we can deduce the expression for

4 bit comparator

$$A = A_3 A_2 A_1 A_0 \quad B = B_3 B_2 B_1 B_0$$

For $A > B$
 if $A_3 > B_3$ or $A_3 = B_3$ and $A_2 > B_2$ or $A_3 = B_3$ and $A_2 = B_2$ and $A_1 > B_1$
 or $A_3 = B_3$ and $A_2 = B_2$ and $A_1 = B_1$ and $A_0 > B_0$

$$A > B = G_3 + E_3 \cdot G_2 + E_3 \cdot E_2 \cdot G_1 + E_3 \cdot E_2 \cdot E_1 \cdot G_0$$

For $A = B \Rightarrow A_3 = B_3$ and $A_2 = B_2$ and $A_1 = B_1$ and $A_0 = B_0$
 $E = E_3 \cdot E_2 \cdot E_1 \cdot E_0$

For $A < B$
 $A_3 < B_3$ or $A_3 = B_3$ and $A_2 < B_2$ or $A_3 = B_3$ and $A_2 = B_2$ and $A_1 < B_1$ or
 $A_3 = B_3$ and $A_2 = B_2$ and $A_1 = B_1$ and $A_0 < B_0$

$$L = L_3 + E_3 \cdot L_2 + E_3 \cdot E_2 \cdot L_1 + E_3 \cdot E_2 \cdot E_1 \cdot L_0$$

In summary: We will start checking with the MSB that if it meets our condition first then we will check what other conditions that may come along considering MSB bits are equal, Means we have to take MSB's are equal and what can be situation of the next lower bits one by one considering the condition we want, and equalizing and check for another next and so on.